Computer

Class 10 (English Medium)

Solved Exercises — Sindh Textbook Board

7 lessons — 157 questions

مسئلو حل کرڻ الگورٿم ڊيزائن کرڻ

[1-1] "To bake a cake" is an example of:

✓ (a) problem

(b) strategy

(c) algorithm

(d) solution

[1-2] To find a feasible solution to a problem, the first step is to:

(a) establish starting point

(b) find available solutions

(c) create a strategy

✓ (d) identify and analyze the problem

[1-3] Step by step solution of a problem in simple language is called:

(a) Problem Solving

✓ (b) Algorithm

(c) Flowchart

(d) Data Structure

[1-4] Which shows the logic of a program graphically?

(a) Data Structure

(b) Graph

(c) Algorithm

✓ (d) Flowchart

[1-5] Which symbol is used for input / output in a flowchart?

(a) Triangle

(b) Square

✓ (c) Parallelogram

(d) Oval

[1-6] Elements of which data structure are not connected sequentially?

(a) Array

✓ (b) Graph

(c) Queue

(d) Stack

[1-7] Which stores data in a hierarchical manner?

(a) Stack

(b) Queue

(c) Array

✓ (d) Tree

[1-8] When data is Pushed in a stack, it means that data is:

✓ (a) inserted

(b) deleted

(c) sorted

(d) edited

[1-9] In a binary tree, each node can have a maximum of:

(a) one node

✓ (b) two nodes

(c) three nodes

(d) four nodes

[1-10] Traversing an array means accessing:

(a) first element

(b) last element

(c) any specific element

✓ (d) each and every element of the array

Question: [1-11] Describe the steps involved in problem solving.

Answer: There are four steps: 1. Define the problem — identifying and describing the real problem; 2. Generate alternative solutions — making a

list of possible solutions; 3. Evaluate and select an alternative — choosing the best solution; 4. Implement and follow up on the solution — measuring the results and seeing whether the solution worked.

Question: [1-12] What are the advantages of developing algorithms?

Answer: An algorithm gives the best possible answer every time; it helps in work needing accuracy and in solving similar problems again and again; computer programs can be built with the help of algorithms; in a good algorithm the input and output are precise, every step is clear, and it can be used in any programming language.

Question: [1-13] List any three advantages of designing flowcharts.

Answer: Flowcharts communicate the logic of a program to others and make it easy to understand; they are useful program documents with which a program can be quickly understood or its logic changed; they work as a guide / blueprint while coding a program; and they show the operations of solving a problem and their sequence visually before writing it.

Question: [1-14] What is the difference between tree and graph data structure?

Answer: A tree shows hierarchical data; it has one root node and every node has one parent with children; a graph has nodes / vertices and edges joining them, no root or child, each node can have many edges, and a cycle can also be formed.

Question: [1-15] What is the difference between queue and stack data structure?

Answer: In a stack, data is added or removed only from the top and it works in LIFO / FILO order (last in, first out); its operations are push and pop. A queue works in FIFO order (first in, first out); a new element enters at the rear end and the old element is removed from the front end; its

operations are enqueue and dequeue.

Question: [1-16] What is the need of index in an array?

Answer: Every element of an array is identified by an index; indexes are a set of consecutive numbers with which any element can be directly found and used. Without an index, it would not be known which element is stored where.

Question: [1-17] With the help of a sketch define: Push, Pop, Overflow and Enqueue, Dequeue.

Answer: Sketch of a stack: with Push a new element enters from the top and with Pop the top element is removed; when the stack is completely full it is in the Overflow state and nothing can be added to it. Sketch of a queue: with Enqueue a new element enters at the rear end and with Dequeue an element is removed from the front end.

Question: [1-18] Design an algorithm to find the greater number by taking two numbers as input.

Answer: Step 1: Start; Step 2: Declare variables num1, num2 and greater; Step 3: Read values num1 and num2; Step 4: If num1 is greater than num2 then greater = num1, else greater = num2; Step 5: Display greater; Step 6: Stop.

Question: [1-19] Design an algorithm to find the area of a triangle.

Answer: Step 1: Start; Step 2: Declare variables base, height and area; Step 3: Read values base and height; Step 4: Apply formula {Area = (base x height) / 2}, i.e. area = (base x height) / 2; Step 5: Display area; Step 6: Stop. [note: the book gives the algorithm pattern; the triangle-area formula is taken from standard maths.]

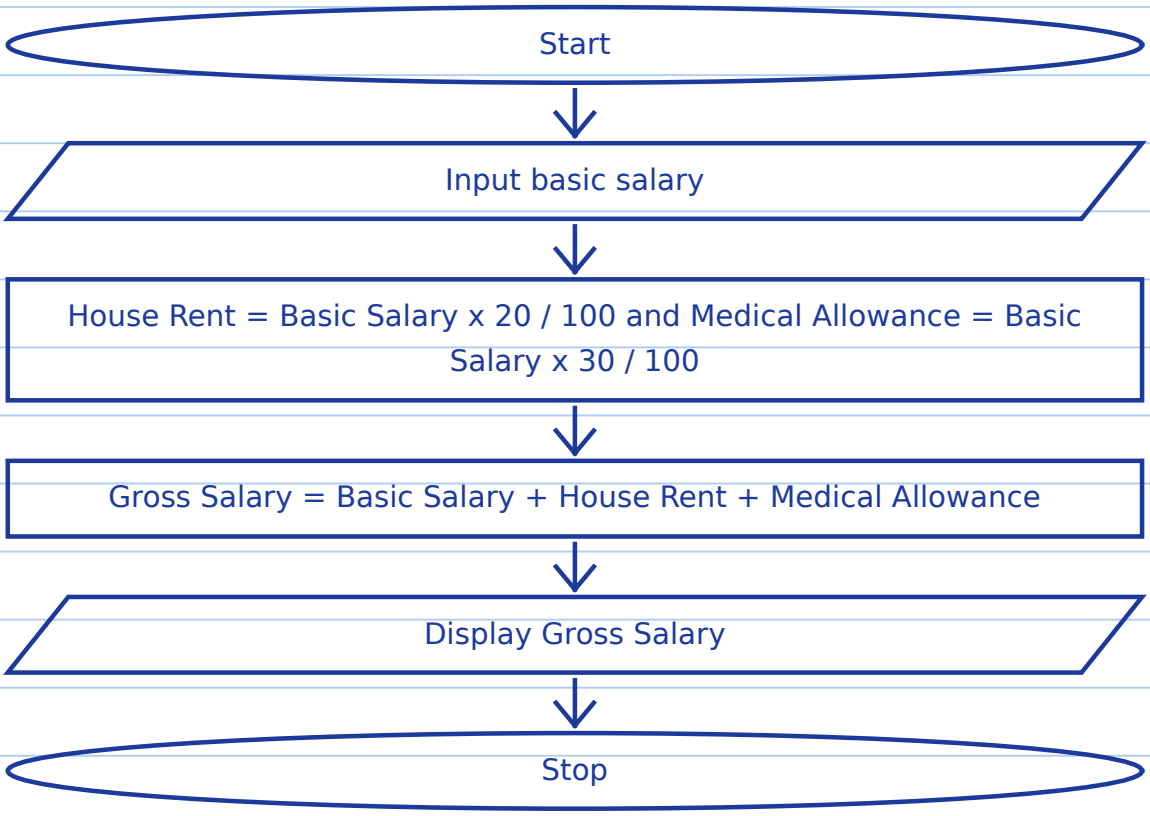
Question: [1-20] Sort the given steps of the algorithm in correct order for baking a cake.

Answer: Correct order: Start; Preheat the oven; Gather the ingredients; Mix

together the ingredients to make the batter; Grease a pan; Pour the batter into the pan; Put the pan in the oven; Set a timer; When the timer goes off, take the pan out of the oven; End.

Question: [1-21] Draw a flow chart to calculate gross salary by adding 20% house rent and 30% medical allowances in basic salary.

Answer: Flowchart: Start (oval) — Input basic salary (parallelogram) — House Rent = Basic Salary x 20 / 100 and Medical Allowance = Basic Salary x 30 / 100 (rectangle) — Gross Salary = Basic Salary + House Rent + Medical Allowance (rectangle) — Display Gross Salary (parallelogram) — Stop (oval). So Gross Salary = 150% of Basic Salary.



Flow diagram drawn from the steps given in the answer.

Question: [1-22] Draw flow charts for all the algorithms given in this unit.

Answer: The flowchart of every algorithm follows the same pattern: Start (oval) — Input (parallelogram) — calculation steps (rectangle) — Display result (parallelogram) — Stop (oval); where a choice is needed, the

decision diamond is used. Making tea, sum of two numbers, average of three numbers, volume of a box, percentage, greater number, area of a triangle and gross salary — each flowchart is made by turning that algorithm's steps into these shapes; the model in the unit (the sum-and-percentage flowchart with the percent > 70 decision) is an example of it.

Question: [1-23] Draw the following structures: (a) Tree with six nodes (b) Graph with five nodes.

Answer: (a) Tree: root node 1 at the top; its children 2 and 3; node 2's children 4 and 5; node 3's child 6 — six nodes in all, every edge from parent to child, no cycle.

(b) Graph: five nodes 1 to 5; edges can join any two nodes, for example 1-2, 2-3, 3-4, 4-5, 5-1 and 1-3 — a cycle can be formed.

سي پلس پلس پروگرامنگ جا بنيادي اصول

[2-1] A computer program is a collection of:

(a) Tasks

✓ (b) Instructions

(c) Computers

(d) Programmers

[2-2] High-level languages have syntax that is:

✓ (a) Easily readable by humans

(b) Easily readable by machines

(c) Easily readable by both

(d) None of the above

[2-3] Low-level languages have syntax that is:

(a) Easily readable by humans

✓ (b) Easily readable by machines

(c) Easily readable by both

(d) None of the above

[2-4] The primary characteristic of a compiler is to:

(a) Translate codes line-by-line

(b) Translate low-level code to machine language

(c) Detect logical errors

✓ (d) Translate codes all at once

[2-5] The primary characteristic of an interpreter is to:

✓ (a) Translate codes line-by-line

(b) Translate low-level code to machine language

(c) Detect logical errors

(d) Translate codes all at once

[2-6] An Integrated Development Environment facilitates a programmer to:

(a) Edit source code

(b) Complete and highlight syntaxes

(c) Debug and compile codes

✓ (d) All of the above

[2-7] All errors, detected by users are typically:

(a) Syntax Errors

(b) Semantic Errors

(c) Run-Time Errors

✓ (d) Logical Errors

[2-8] Allowed names for declaring a variable:

(a) Can contain whitespaces

(b) Can be one of the reserved words

✓ (c) Can contain letters, digits and underscores

(d) Can be the same as its data type

[2-9] A bool data can store following type of value:

(a) Numbers

(b) Strings

(c) Fractional numbers

✓ (d) True or false

[2-10] Which data type occupies the most space in memory?

(a) Character

(b) Integer

(c) Floating point

✓ (d) Double floating point

Question: [2-11] What is computer program?

Answer: A computer program is a set of instructions that is understood by

a computer to perform tasks. The person who writes programs is called a programmer, and specialized programs such as a compiler convert those instructions into machine-readable instructions.

Question: [2-12] List five common high-level languages used and describe their purpose.

Answer: Languages named in the unit: C++, Java, Python, Visual Basic and PHP. C++ is a general-purpose high-level language developed as an enhancement of the C language to include the object-oriented concept and to make writing programs easier and more pleasant. [note: the book does not state a separate purpose for each of these five languages; separate purposes for the other four are not given in the book.]

Question: [2-13] Using the rules of naming variable, develop ten meaningful and valid variable names.

Answer: Ten valid names under the rules: studentName, totalMarks, firstName, lastName, rollNo, _average, myVar1, age2, price, class10 — all begin with a letter or an underscore, contain letters, digits and underscores, have no whitespace, use no reserved word, and are all shorter than 32 characters.

Question: [2-14] Write any two differences between machine and assembly language.

Answer: Machine language is at the lowest level, written in 0 and 1 (binary), executed directly by the computer, and needs no translator; it is the first-generation language. Assembly language is one level above machine language, written in simple English language so it is easy for humans to read, write and maintain, and it needs an assembler to convert it into machine code; it is the second-generation language.

Question: [2-15] What are Strings in C++?

Answer: Strings are variables that store non-numerical values longer than

one single character. The C++ language library supports strings through the standard string class; a string is not a fundamental type but behaves in a similar way to fundamental types in its most basic use; it can be declared without an initial value and assigned a value during execution.

Question: [2-16] What is the difference between declaring and initializing a variable?

Answer: A declaration tells the compiler where and how much storage to create for the variable; it specifies a data type and a name: data_type variable_name;. Initialization assigns an initial value in the declaration itself, with an equal sign: data_type variable_name = value;. Variables do not need to be assigned initial values; a value can also be assigned later.

Question: [2-17] What is the difference between source code and object code?

Answer: Source code is a program written in any language, as the programmer writes it. Object / machine code is the translated form that the computer runs; the translator converts the source code into the target machine language. [note: the book uses the words "source code" and "machine code / target machine language", not the term "object code"; the answer follows from its discussion of translators.]

Question: [2-18] List any four advantages of using an IDE.

Answer: An IDE gives compilation, linking, loading and debugging tools in a single environment; code completion improves the programming workflow; it automatically checks for errors to ensure top-quality code; refactoring allows comprehensive and mistake-free renaming; and it also gives facilities such as syntax highlighting and autocomplete.

سي پلس پلس ان پٹ / آئوٹ پٹ سنپالٹ

[3-1] The C++ header file ___ contains function prototype for the standard Input and Output functions.

(a) <iomain.h>

✓ (b) <iostream>

(c) <fstream.h>

(d) <cstdio.h>

[3-2] Which operator is used for input stream?

(a) >

(b) <<

✓ (c) >>

(d) <

[3-3] gets stands for ___.

(a) get stream

✓ (b) get string

(c) get str

(d) get std

[3-4] getch() and getche() are included in ___ header file.

(a) <cstdio.h>

✓ (b) <conio.h>

(c) <stdlib.h>

(d) <stdio.h>

[3-5] Which operator is used for logical AND operation?

(a) &

✓ (b) &&

(c) ||

(d) !

[3-6] Which of the following operator is correct to compare two values of variables?

(a) =

(b) <=

(c) ==

✓ (d) both b and c

[3-7] Which of the following needs pressing Enter Key from the keyboard?

[3-8] != operator belong to which type of operator.

✓ (a) Relational

(b) Logical

(c) Arithmetic

(d) None of these

[3-9] Which operator add the first operand to the second operand and gives the result to first operand.

(a) *=

✓ (b) +=

(c) ++

(d) +

[3-10] cout << 12-6/2; What will be the result on screen?

(a) 3

(b) 6

✓ (c) 9

(d) 12

Question: [3-11] Use \a and \r both escape sequences in a program.

Answer: #include<iostream> using namespace std; int main() { cout << "Beep\a"; cout << "\rStart"; return 0; } — \a is Alert / alarm, it causes a beep sound in the computer; \r is Carriage Return, it positions the cursor

at the beginning of the current line.

Question: [3-12] How many types of comment statements are used in C++?

Answer: Two types: 1. Single Line Comment — with a double slash (//), for a one-line explanation; 2. Multi Line Comment — with (/* and */), for explanations over many lines. Both are ignored by the compiler, meaning they are not executed.

Question: [3-13] Differentiate between Arithmetic operators and Relational operators.

Answer: Arithmetic operators (+, -, *, /, %) perform arithmetic operations / calculations on operands and give a numeric result. Relational operators (==, !=, >, <, >=, <=) test the relation between two values; they are binary operators, they need two operands, and their result is True (1) or False (0).

Question: [3-14] Write a program in C++ and use all arithmetic assignment operators.

Answer: #include<iostream> using namespace std; int main() { int a = 20; a += 2; cout << "After += 2, a = " << a << endl; a -= 3; cout << "After -= 3, a = " << a << endl; a *= 4; cout << "After *= 4, a = " << a << endl; a /= 4; cout << "After /= 4, a = " << a << endl; return 0; } — Output: After += 2, a = 22; After -= 3, a = 19; After *= 4, a = 76; After /= 4, a = 19.

Question: [3-15] What is basic difference between Assignment operator and Equal to operator.

Answer: The assignment operator (=) assigns the value of the right-side expression to the left-side variable, for example x=10;. The equal to operator (==) checks the equality of the values of two operands, for example x==y; if the condition is true the result is true, otherwise false.

Question: [3-16] What is the basic difference between \n and \t?

Answer: \n means New line / line feed: it inserts a new line and the cursor moves to the beginning of the next line. \t means Horizontal tab: it shifts the cursor a couple of spaces to the right in the same line.

Question: [3-17] Get the output of following program: int a = 27; then a is displayed with a++, a, --a and a.

Answer: Output: a is 27; a is now 27; a is now 28; a is now 27; a is now 27. Reason: a++ is postfix, first 27 is printed then a becomes 28; then a is printed as 28; --a is prefix, first a becomes 27 then 27 is printed; finally a is printed as 27.

Question: [3-18] Develop programs for manipulating the following formulas: Speed s=d/t; Newton's second law of motion F=ma; Acceleration a=(vf-vi)/t; Area of Triangle a=1/2 bh; Convert the Celsius to Fahrenheit F=(c*1.8)+32.

Answer: 1. Speed: input d and t; s = d / t; display s. 2. Force: input m and a; F = m * a; display F. 3. Acceleration: input vf, vi and t; a = (vf - vi) / t; display a. 4. Area of triangle: input b and h; area = (b * h) / 2; display area. 5. Celsius to Fahrenheit: input c; F = (c * 1.8) + 32; display F. Every program has the same structure: #include<iostream>, using namespace std;, int main(), declaration of variables, input with cin, formula, result with cout, return 0;.

Question: [3-19] Write a program to calculate the volume of a box.

Answer: #include<iostream> using namespace std; int main() { float length, width, height, volume; cout << "Enter length, width and height"; cin >> length >> width >> height; volume = length * width * height; cout << "Volume = " << volume; return 0; } — Volume = length x width x height (the formula of Algorithm 4 of Unit 1).

Question: [3-20] Write a program of Marksheet takes input of five subjects,

print its total and percentage also.

Answer: #include<iostream> using namespace std; int main() { float m1, m2, m3, m4, m5, total, percentage; cout << "Enter marks of five subjects"; cin >> m1 >> m2 >> m3 >> m4 >> m5; total = m1 + m2 + m3 + m4 + m5; percentage = total / 500 * 100; cout << "Total = " << total << " Percentage = " << percentage; return 0; } — Total 500 because each subject is of 100 marks (like the book's three-number model: Percent = Sum / 300 * 100).

Question: [3-21] Write a code to calculate mathematical expression of a2+2ab+b2.

Answer: #include<iostream> using namespace std; int main() { float a, b, result; cout << "Enter a and b"; cin >> a >> b; result = a * a + 2 * a * b + b * b; cout << "Result = " << result; return 0; } — this expression equals the square of (a + b).

Question: [3-22] List out errors from the C++ program and remove those errors, write the output.

Answer: Errors: 1. using namespace std is missing a semicolon (;) at the end; 2. int main() must not be followed by (;); 3. cin >> x is missing (;) at the end; 4. in a * a, a is not declared — it should be x * x. Correct program: #include<iostream> using namespace std; int main() { int x; cout << "Enter the value of x"; cin >> x; cout << "The square of x is " << x * x; return 0; } — Output (for input 5): Enter the value of x 5; The square of x is 25.

کنٹرول اسٹرکچر

[4-1] Loop within a loop is called ___ loop.

(a) inner

(b) outer

(c) enclosed

✓ (d) nested

[4-2] "case" and ___ are also part of "switch" statement.

(a) have

✓ (b) default

(c) for

(d) if

[4-3] "for" Loop expression has ___ parts.

(a) one

(b) two

✓ (c) three

(d) four

[4-4] exit() function is used to ___.

(a) close function

(b) close loop

✓ (c) close program

(d) close switch

[4-5] "continue" statement takes control to the ___.

✓ (a) top of loop

(b) end of loop

(c) top of function

(d) end of function

[4-6] In "goto" statement label is followed by ___ character.

[4-7] To send value to the calling function we use ___ statement.

(a) throw

✓ (b) return

(c) send

(d) back

[4-8] "break" statement is used with ___.

(a) if

✓ (b) switch

(c) for

(d) while

[4-9] Using "else" is ___ with "if" statement.

(a) prohibited

(b) advised

(c) compulsory

✓ (d) optional

[4-10] "if" and loop expressions use ___ operators to test condition.

(a) arithmetic

✓ (b) relational

(c) insertion

(d) bitwise

Question: [4-11] What is the purpose of "default" statement in C++?

Answer: In a switch, if the switch variable does not match any of the case constants, control goes to default (if it is present). Default acts as "else".

Question: [4-12] Can we use "while" loop in place of "for" loop, if yes then how?

Answer: Yes. The three parts of a for loop can be split up: initialization before the loop, the condition in while(...), and increment / decrement

inside the body of the loop. For example: int i = 1; while (i <= 10) { cout << i; i++; } — this is the same as for (i = 1; i <= 10; i++).

Question: [4-13] What is the main difference between while and do while loops?

Answer: while is a pre-test loop: the condition is tested at the start of the loop, so the body may not run even once. do while is a post-test loop: the condition is tested at the end of the body of the loop, so its body runs at least once; its while(condition) is followed by a semicolon.

Question: [4-14] Write the function of for loop.

Answer: The for loop executes a sequence of statements multiple times. It is usually used in situations where we know at the start of the loop how many times the loop block will execute; it is a counter-controlled / definite repetition loop and the condition is tested at the start of the loop.

Question: [4-15] Why we make block of statements using braces?

Answer: To gather more than one statement into one group / block, they are enclosed in curly braces, so that they run together as one joint part; for example, if there is more than one statement in an if block, the body of a loop or the body of a function, they are all enclosed in braces.

Question: [4-16] Which data type variables can be used in "switch" statement?

Answer: In a switch, only character and integer data type variables can be used; it matches the switch variable's value only with case constants, not with any variable, and relational operators like < or > cannot be used in it — the value must exactly match the case constants.

Question: [4-17] What is the purpose of jump statements?

Answer: Jump statements change the execution of the program from its normal sequence: they transfer control to another part of the program, terminate a loop or switch, send a value back from a function, or

terminate the program.

Question: [4-18] Write the purpose of following statements: (a) else if (b) switch (c) goto (d) exit.

Answer: (a) else if: tests many conditions in a ladder — the block of whichever condition is true runs, otherwise the next condition is tested; it makes a nested-if program simpler and more readable.

(b) switch: matches the variable's value with the case constants and transfers control to the matching case; if none matches, to default.

(c) goto: jumps control unconditionally from the goto to a labeled statement in the same function.

(d) exit: terminates the C++ program and closes it whenever executed.

Question: [4-19] Match the column: (a) if — which one from column B matches it?

Answer: (vi) else — if and else together make the if-else pair.

Question: [4-20] Match the column: (b) loop — which one from column B matches it?

Answer: (v) iteration — a loop means running statements again and again, i.e. iteration.

Question: [4-21] Match the column: (c) Conditional expression — which one from column B matches it?

Answer: (i) Relational operators — conditional expressions are tested with relational operators.

Question: [4-22] Match the column: (d) Loops and switch — which one from column B matches it?

Answer: (ii) break — break terminates the loop or switch statement.

Question: [4-23] Match the column: (e) do — which one from column B matches it?

Answer: (vii) while — do and while together make the do-while loop.

Question: [4-24] Match the column: (f) >> — which one from column B matches it?

Answer: (iv) operator — >> is an operator (the extraction / input operator).

Question: [4-25] Match the column: (g) case — which one from column B matches it?

Answer: (iii) switch — case is a part of the switch statement.

Question: [4-26] Write a program that takes a number as input and print whether it is odd or even.

Answer: #include<iostream> using namespace std; int main() { int num; cout << "Enter a number"; cin >> num; if (num % 2 == 0) cout << "Even"; else cout << "Odd"; return 0; } — if dividing the number by 2 leaves remainder 0, it is even, otherwise odd.

Question: [4-27] Write a program to add numbers from 1 to 20.

Answer: #include<iostream> using namespace std; int main() { int i, sum = 0; for (i = 1; i <= 20; i++) sum = sum + i; cout << "Sum = " << sum; return 0; } — Output: Sum = 210.

Question: [4-28] Write a program that take month number as input (from 1 to 12) and print number of days in that month. If wrong number is given then show error message.

Answer: #include<iostream> using namespace std; int main() { int month; cout << "Enter month number 1 to 12"; cin >> month; switch (month) { case 1: case 3: case 5: case 7: case 8: case 10: case 12: cout << "31 days"; break; case 4: case 6: case 9: case 11: cout << "30 days"; break; case 2: cout << "28 days"; break; default: cout << "Wrong number

/ Error message"; } return 0; } — [note: February is written as 28 days; the book does not give the leap-year case.]

Question: [4-29] Input a number up to six digits and show each digit in separate line.

Answer: #include<iostream> using namespace std; int main() { int num, divisor = 1, temp; cout << "Enter a number up to six digits"; cin >> num; temp = num; while (temp >= 10) { divisor = divisor * 10; temp = temp / 10; } while (divisor > 0) { cout << num / divisor << endl; num = num % divisor; divisor = divisor / 10; } return 0; } — the first loop finds the divisor of the largest place, the second loop prints each digit from left to right on a separate line.

Question: [4-30] Take input a character, number of rows and number of columns. Draw a square box filled with that character with given number of rows and columns.

Answer: #include<iostream> using namespace std; int main() { char ch; int rows, cols, i, j; cout << "Enter character, rows and columns"; cin >> ch >> rows >> cols; for (i = 1; i <= rows; i++) { for (j = 1; j <= cols; j++) cout << ch; cout << endl; } return 0; } — the outer loop runs the rows, the inner loop prints the character in each row.

Question: [4-31] Write a programs that generate the following outputs: (1) * / ** / *** / **** / ***** (2) 1 / 12 / 123 / 1234 / 12345 (3) 1 / 22 / 333 / 4444 / 55555.

Answer: (1) stars: for (i = 1; i <= 5; i++) { for (j = 1; j <= i; j++) cout << "*"; cout << endl; } (2) numbers: for (i = 1; i <= 5; i++) { for (j = 1; j <= i; j++) cout << j; cout << endl; } (3) same number: for (i = 1; i <= 5; i++) { for (j = 1; j <= i; j++) cout << i; cout << endl; } — all three are made with nested loops, the outer loop runs the row, the inner loop runs the items of each row.

Question: [4-32] Write a program that takes a number as input and print whether it is prime or not.

Answer: #include<iostream> using namespace std; int main() { int n, i; bool isPrime = true; cout << "Enter a number"; cin >> n; if (n <= 1) isPrime = false; for (i = 2; i <= n / 2; i++) { if (n % i == 0) isPrime = false; } if (isPrime) cout << "Prime"; else cout << "Not Prime"; return 0; } — a prime number is one that is divided only by 1 and itself.

Question: [4-33] Take salary as input and on its basis show different levels of designations in an organization like manager, supervisor, worker etc.

Answer: #include<iostream> using namespace std; int main() { float salary; cout << "Enter salary"; cin >> salary; if (salary >= 100000) cout << "Manager"; else if (salary >= 50000) cout << "Supervisor"; else cout << "Worker"; return 0; } — [note: the book does not fix salary limits; the limits above are given as an example.]

Question: [4-34] Write a program that prints square of numbers from 1 to 10.

Answer: #include<iostream> using namespace std; int main() { int i; for (i = 1; i <= 10; i++) cout << i << " square = " << i * i << endl; return 0; } — Output: 1 square = 1, 2 square = 4, 3 square = 9, 4 square = 16, 5 square = 25, 6 square = 36, 7 square = 49, 8 square = 64, 9 square = 81, 10 square = 100.

Question: [4-35] Take a number and print its table from 1 to 10 using while loop according to the following format.

Answer: #include<iostream> using namespace std; int main() { int n, i = 1; cout << "Enter a number"; cin >> n; while (i <= 10) { cout << n << " x " << i << " = " << n * i << endl; i++; } return 0; } — for example, for 2 the output is: 2 x 1 = 2, 2 x 2 = 4, 2 x 3 = 6, ..., 2 x 10 = 20.

فنکشن

Question: [5-1] The functions that are defined by the programmer are called: (a) Built-In function (b) User-defined-function (c) Subfunction (d) Function

Answer: (b) User-defined-function — the programmer can also write their own functions to perform a specific task; they are called user-define-functions.

[5-2] A programmer creates a function for a particular task and the programmer wants to include that function in program. Which extension is required to save that function?

(a) .obj

✓ (b) .h

(c) .cpp

(d) .exe

[5-3] In C++, int main() returns which data type value by default?

(a) float

✓ (b) int

(c) char

(d) double

[5-4] The parameters specified is the function header are called:

✓ (a) formal parameters

(b) actual parameters

(c) default parameters

(d) command line parameters

[5-5] The word "prototype" means:

✓ (a) Declaration

(b) Calling

(c) Definition

(d) Both a & b

[5-6] The function prototype consists of:

(a) Name of function

(b) the parameters are passed to the function

(c) The value return from function

✓ (d) All of these

[5-7] All variables declared in function definition are called

✓ (a) Local variable

(b) Instance variable

(c) Global variable

(d) Static variable

[5-8] Which are not the built-In function?

Question: [5-9] Differentiate between function declaration and function definition.

Answer: Declaration: a function without its definition (code block); declared before main(); tells the compiler about the function's name, return data type, and arguments/parameter data types; ends with a semicolon (;). Definition: the function itself; it has a header and a body / code block; the header has the return value data type, the function name and the list of arguments with datatypes in parenthesis; the body has statements in braces; the definition may be before or after the main function.

Question: [5-10] What is the purpose of keyword "void" in function?

Answer: void is used when no value is returned by the function. If there is no parameter in the function, void is also used, for example void function name(void);.

Question: [5-11] Why we use header files?

Answer: Predefined functions are declared in header files; to use them the proper header file is included with #include, for example cmath for sqrt(), cstdio for gets() / puts(), conio.h for getch() / getche(). Header files link the external header files / libraries which may be required in the program.

Question: [5-12] Differentiate between passing argument and return the value from function.

Answer: Passing an argument: when the function is called, actual values (variables or constants) are given with the function call; they are received in the formal parameters of the function definition, which act as local variables inside the function. Returning a value: when the function completes its execution, the return keyword sends a single value back to the calling function; the return data type must be specified in the header, before the function name, in both the function declaration and the definition.

Question: [5-13] What is the difference between external variables and local variables?

Answer: External / global variables are declared outside of the main function; they are also known as external variables; they hold the value of the variable during the entire execution of the program and their value can be shared with different functions. Local variables are declared inside any function (also in the header of the function definition) and are only accessible within a specific part of the program / that function.

Question: [5-14] List the five standard built-In functions with examples.

Answer: 1. sqrt(a) — finds the square root of a number (the book's example: the square root of a with sqrt(a)). 2. pow(x, y) — finds x to the power y. 3. strcpy(str1, str2) — copies one string into another string. 4. toupper(ch) — converts a character to upper case. 5. getch() — gets a

single character from the keyboard. [note: the book gives the names of these functions and shows a worked example only for sqrt(); the other examples follow the same calling pattern.]

Question: [5-15] Write down the advantages of User-define functions in C++.

Answer: 1. Large and complicated programs are broken down into smaller modules / functions. 2. Functions help the programmer to manage the code of the large program. 3. A function written once can be called from any part of the program. 4. The programmer can create functions for a specific task. [note: the book does not give a separate list of advantages; these advantages follow from its introduction (breaking the program into smaller modules, managing the code of a large program).]

Question: [5-16] Get the output and highlight the errors from the following program.

Answer: Error: in the function prototype line, \\ is written, which is not a comment in C++; a comment needs //: void Table(void); // Function prototype. There is no other error. Output (for input m = 5): The value of m..... 5; then the table of 5: 5*1=5, 5*2=10, 5*3=15, 5*4=20, 5*5=25, 5*6=30, 5*7=35, 5*8=40, 5*9=45, 5*10=50.

Question: [5-17] Write a program on the following given series by using For loop. 0, 5, 10, 15, 20, 25, 30, 35. Apply the technique (no return value and no pass parameters) in program.

Answer: #include<iostream> using namespace std; void series(void); int main() { series(); return 0; } void series(void) { int i; for (i = 0; i <= 35; i = i + 5) cout << i << " "; } — Output: 0 5 10 15 20 25 30 35. The function neither returns a value nor takes parameters: void function name(void);.

Question: [5-18] Write a program to take input from the keyboard and check whether given number is Even or Odd. Apply the technique (return

value and pass parameters) in program.

Answer: #include<iostream> using namespace std; int checkEven(int n); int main() { int num, result; cout << "Enter a number"; cin >> num; result = checkEven(num); if (result == 1) cout << "Even"; else cout << "Odd"; return 0; } int checkEven(int n) { if (n % 2 == 0) return 1; else return 0; } — the function takes a parameter and returns a value: int function name(int);.

Question: [5-19] Write a program to convert kilogram in grams using function. The function should take value in kilogram as parameter and should return value in grams.

Answer: #include<iostream> using namespace std; float kgToGrams(float kg); int main() { float kg, grams; cout << "Enter kilograms"; cin >> kg; grams = kgToGrams(kg); cout << "Grams = " << grams; return 0; } float kgToGrams(float kg) { return kg * 1000; } — [note: the book does not state that 1 kilogram = 1000 grams; this is the standard metric conversion.]

Question: [5-20] Create a function that takes length and height as arguments and print a box of stars accordingly. e.g. length =10, height = 4. Apply the technique (return value but not pass parameters) in program.

Answer: #include<iostream> using namespace std; int starBox(int length, int height); int main() { int total; total = starBox(10, 4); cout << "Total stars = " << total; return 0; } int starBox(int length, int height) { int i, j; for (i = 1; i <= height; i++) { for (j = 1; j <= length; j++) cout << "*"; cout << endl; } return length * height; } — Output: 4 rows of 10 stars, then Total stars = 40. [note: the two parts of the exercise conflict — it asks to take length and height as arguments and also says "not pass parameters"; the program takes the arguments and also returns a value.]

Question: [5-21] Write a function that returns factorial of a given number.

Answer: #include<iostream> using namespace std; int factorial(int n); int main() { int num; cout << "Enter a number"; cin >> num; cout << "Factorial = " << factorial(num); return 0; } int factorial(int n) { int i, f = 1; for (i = 1; i <= n; i++) f = f * i; return f; } — [note: the book does not define factorial; the standard meaning is used: n! = 1 x 2 x ... x n.]

Question: [5-22] Develop programs for manipulating the following formulas in form of function: Area of Rectangle a = l b; Area of Circle a = π r2 (pi = 3.14); Pythagorean Theorem c2 = a2 + b2.

Answer: 1. Rectangle: float rectArea(float l, float w) { return l * w; } — area = length x width. 2. Circle: float circleArea(float r) { return 3.14 * r * r; } — area = pi x radius x radius. 3. Pythagoras: float hypotenuse(float a, float b) { return sqrt(a * a + b * b); } — from the given formula c2 = a2 + b2, c = sqrt(a2 + b2). Each function is called from main() with parameters and returns the result.

ڈیجیٹل لاجک ڈیزائن

[6-1] The universal gate is ___.

✓ (a) NAND Gate

(b) AND Gate

(c) OR Gate

(d) None of these

[6-2] The ___ is Inverter.

(a) AND

(b) OR Gate

✓ (c) NOT

(d) None of these

[6-3] In Boolean Algebra, the bar sign (-) indicates ___.

(a) OR Operation

(b) NOR Operation

✓ (c) NOT Operation

(d) Both b and c

[6-4] The Boolean Algebra is used for ___.

(a) Creating Circuits

(b) Apply 12 rules of Boolean

✓ (c) Simplify the Boolean expression

(d) Differentiate the gates

[6-5] With the combination of three variables, how many outputs are expected altogether?

(a) Three

(b) Six

✓ (c) Eight

(d) Nine

[6-6] A+A=A is a ___ rule of Boolean Algebra.

(a) 3rd

(b) 6th

✓ (c) 5th

(d) 7th

[6-7] A.Ā=0 is a ___ rule of Boolean Algebra.

(a) 1st

✓ (b) 8th

(c) 6th

(d) 10th

[6-8] Simply form of Boolean expression of ABC + ABĈ + ĀB is ___.

(a) A

✓ (b) B

(c) C

(d) B̄

[6-9] Simplify form of Boolean expression of ABC+ĀBC is ___.

(a) AC

✓ (b) BC

(c) B

(d) Ā

Question: [6-10] Boolean expression of the given circuit is ___. (A and B go into an AND gate, its output passes through a NOT gate.)

Answer: (d) (A.B) with a bar over it — the AND gate's output is A.B and the NOT gate inverts it, so the circuit is a NAND gate: Y = (A.B) with a bar over it.

Question: [6-11] Explain all basic logic gates and their operations.

Answer: AND gate: two or more inputs, a single output; operates on logical

multiplication; Y = A.B; output is 1 only when all inputs are 1, if any input is 0 the output is 0. OR gate: two or more inputs, a single output; operates on logical addition; Y = A+B; output is 1 if any input is 1, if all inputs are 0 the output is 0. NOT gate: a single input, a single output; inverter; Y = A with a bar over it; the output is the inversion of the input.

Question: [6-12] Differentiate between NAND and NOR gates.

Answer: NAND gate (NOT-AND): the collection of an AND gate and a NOT gate connected in series; Y = (A.B) with a bar over it; the output is 0 only when all inputs are 1, if any one of the inputs is 0 the output is 1. NOR gate: the collection of an OR gate and a NOT gate; the output is inverted OR; Y = (A+B) with a bar over it; the output is 1 only when both inputs are 0, if any one of the inputs is 1 the output is 0. Both are universal gates.

Question: [6-13] Why do we use Boolean Algebra?

Answer: Boolean algebra is used to describe the logical operations and processing of binary information; its pre-defined rules help to simplify the logical expression, so that the number of logic gates needed to perform a particular logic operation is reduced.

Question: [6-14] Explain the function of Inverter.

Answer: The inverter (NOT gate) inverts the input: if the input is 1 the output is 0, if the input is 0 the output is 1; Y = A with a bar over it.

Question: [6-15] Explain the purpose of Truth table.

Answer: A truth table is a tabular representation of all the combinations of values for inputs and their corresponding outputs.

Question: [6-16] Simplify the following Boolean expression. Z = AB + A(B + C) + B(B + C).

Answer: Expand: AB + AB + AC + BB + BC = AB + AC + B + BC (because BB = B, Rule 7, and AB + AB = AB, Rule 5). B + BC = B (Rule 10: A + A.B = A). Then Z = AB + AC + B = B + AB + AC = B(1 + A) + AC = B + AC

(Rule 2: 1 + A = 1; Rule 4: B.1 = B). So Z = B + AC.

Question: [6-17] Draw a Logic Circuit of 10th Law of Boolean Algebra.

Answer: 10th Rule: A + A.B = A. Circuit: A and B go into an AND gate to give A.B; then A and A.B go into an OR gate to give A + A.B, whose output is A.

Question: [6-18] Design a Logic Circuit from Boolean expression Q = (A.B) + (A.B) + (A+B) with a bar over (A+B).

Answer: Circuit: two AND gates (both with A and B) give the two A.B parts; one OR gate (with A and B) followed by a NOT gate gives (A+B) with a bar over it; then the three parts go into a three-input OR gate, whose output is Q. [note: A.B is printed twice in the expression; AB + AB = AB (Rule 5), so Q = AB + (A+B) with a bar over (A+B).]

Question: [6-19] Derive the Boolean expression from the given circuit and make a truth table and simplify that Boolean expression. (Circuit: A, B, C into a three-input AND gate; B and C through NOT gates into an OR gate; A and (B-bar + C-bar) into an AND gate; both results into a final OR gate.)

Answer: Expression: Y = ABC + A(B-bar + C-bar). Truth table (A B C | ABC | B-bar+C-bar | A(B-bar+C-bar) | Y): 0 0 0 | 0 | 1 | 0 | 0; 0 0 1 | 0 | 1 | 0 | 0; 0 1 0 | 0 | 1 | 0 | 0; 0 1 1 | 0 | 0 | 0 | 0; 1 0 0 | 0 | 1 | 1 | 1; 1 0 1 | 0 | 1 | 1 | 1; 1 1 0 | 0 | 1 | 1 | 1; 1 1 1 | 1 | 0 | 0 | 1. Simplify: Y = A(BC + B-bar + C-bar); B-bar + BC = B-bar + C (Rule 11); so Y = A(B-bar + C + C-bar) = A(B-bar + 1) = A (Rule 6 and Rule 2 and Rule 4). So the simplified form is: Y = A.

اسکریچ جو تعارف

[7-1] The feature of Scratch is ___.

(a) It is a visual

(b) Its free forever

(c) No need to remember codes

✓ (d) All of above

[7-2] In Scratch, the character which moves on the Stage is called a ___.

✓ (a) Sprite

(b) Command

(c) Script

(d) Event

[7-3] Repeat 10, forever and if ... then codes are available in ___.

(a) Motion

✓ (b) Control

(c) Look

(d) Sensing

[7-4] The Looks of Sprite can be changed by using ___.

(a) Backdrop Tab

✓ (b) Costume Tab

(c) Script Tab

(d) Control Tab

[7-5] A Scratch Program is at least made up with ___.

(a) Many Sprites

✓ (b) One Sprite and Code

(c) One Sprite

(d) One Sprite and Backdrop

[7-6] To change the position of Sprite on screen, we use ___.

✓ (a) Coordinates

(b) Stage Information

(c) Command Pallets

(d) Costume Tab

[7-7] "move 10 steps" can be found under ___.

(a) Look

✓ (b) Motion

(c) Sound

(d) Control

[7-8] Turn command, turns Sprite to specified:

(a) Coordinate

(b) Steps

✓ (c) Degree

(d) Seconds

[7-9] This repeat 10 command is available in ___.

(a) Event

✓ (b) Control

(c) Motion

(d) Looks

Question: [7-10] In the given picture identify the x and y coordinates (positions) of Point - A.

Answer: (a) x=100, y=-100 — Point A is in the lower-right quadrant, where x is positive and y is negative.

Question: [7-11] Explain the following: (i) Script (ii) Sprite (iii) Backdrop.

Answer: (i) Script: the set of step-wise instructions given to the sprite to do a particular task; scripts are instructions that make sprites perform a task; each sprite has its own script area.

(ii) Sprite: the images of cartoons, characters or objects added in the project; we can have multiple sprites in a project but at least one sprite is always needed; Cat is the default sprite.

(iii) Backdrop: the background added on the stage; by default, no backdrop is added in a project; the look of the stage can be changed by adding new backdrops.

Question: [7-12] State the difference between repeat 10 and forever commands.

Answer: repeat 10 repeats the following code blocks for the specified number of times (10 times). forever keeps repeating the following code blocks for infinite times.

Question: [7-13] Write the use of the following codes: forever, wait, say, play sound, go to x, y.

Answer: forever: keeps repeating the following code blocks for infinite times. wait: waits for specified seconds. say: says the given words / phrase (say ... for 2 secs says it for the specified seconds). play sound: plays the specified sound. go to x, y: sends the sprite to the specified x, y coordinates (position on stage).

Question: [7-14] What is the difference between using Scratch online and offline?

Answer: Online: the editor is loaded in the web browser at scratch.mit.edu/projects/editor and used; it functions just like the offline editor; an account can be created on Scratch to save projects online. Offline: Scratch is downloaded and installed from the MIT website on a computer or an Android device to work offline (run the exe file, choose the location, shortcuts).

Question: [7-15] In your notebook mark the color of each pallet in Script

Tab. One is done for you. How are these colors helpful for the user?

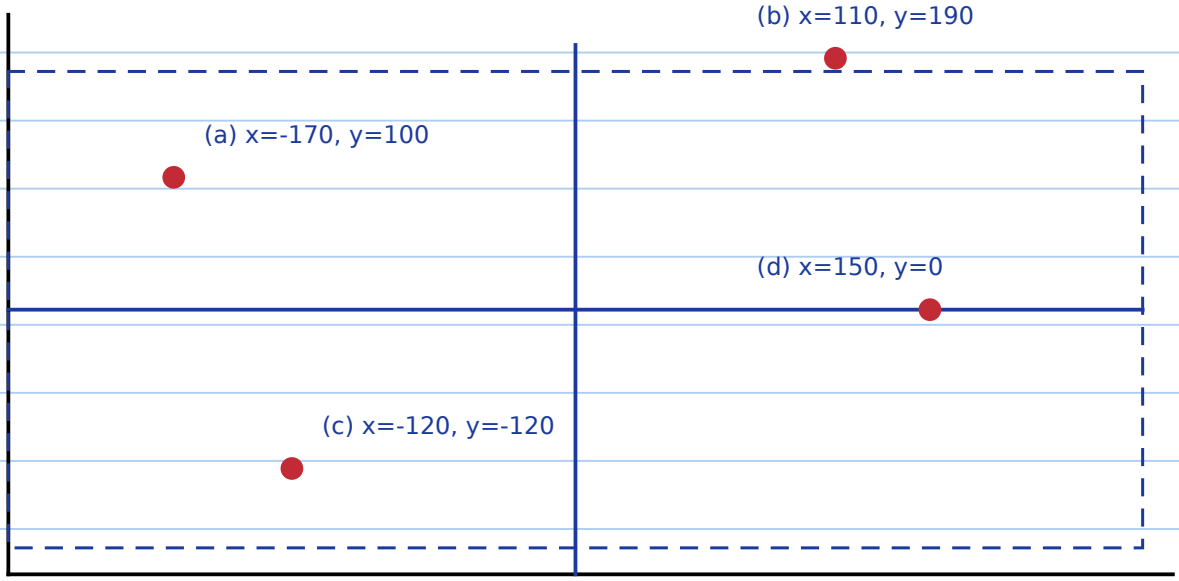
Answer: Each block group has its own associated color (in the book's figures: Motion blue, Events brown / orange, Sound pink, Looks purple, Control golden yellow, Sensing light blue, Data orange, Pen green, Operators green). The colors differentiate the different commands and help to quickly find the right block / option — the book says the colors will help you to find the option.

Question: [7-16] Draw a matrix of 480 x 360 and then mark these points on the that matrix: (a) x=-170, y=100 (b) x=110, y=190 (c) x=-120, y=-120 (d) x=150, y=0. Now with the help of go to x-y command, check your marks on the matrix.

Answer: (a) x=-170, y=100: upper-left (second) quadrant — x negative, y positive.

(b) x=110, y=190: upper-right (first) quadrant — x positive, y positive. [note: the stage's y range is -180 to 180, so y=190 lies slightly outside the top edge.] (c) x=-120, y=-120: lower-left (third) quadrant — x negative, y negative.

(d) x=150, y=0: on the x-axis (the middle horizontal line), on the right side. To check: for each point use the go to x: ... y: ... block and see whether the sprite goes to the marked place or not.



Plotted from the values given in the question/answer.

Question: [7-17] Make a sample program in which two Sprites talk with each other. Try making few jokes.

Answer: Sample program (the conversation lines are a sample): when flag clicked: Sprite 1: say "Hello! How are you?" for 2 secs; Sprite 2: say "I am fine! What about you?" for 2 secs; Sprite 1: say "I am fine too!" for 2 secs; Sprite 2: say "Why did the computer go to school? To improve its memory!" for 2 secs. Each sprite has its own code block and the turn changes with a wait block between the say blocks.

Question: [7-18] Develop a program to enter Radius of the Circle; calculate area and circumference of the circle and display the result.

Answer: When flag clicked: ask "Enter radius" and wait; set radius to answer; set area to 3.14 x radius x radius; set circumference to 2 x 3.14 x radius; say (join "Area = " area); say (join "Circumference = " circumference). [note: the book gives the circle area formula in Unit 5 (pi r2, pi = 3.14); the circumference formula 2 pi r is not given in the book.]